

DAY 4 – DATA VISUALISATION & POWER BI

Beginner Training Notes • Matplotlib • Seaborn • Power BI

Objective: Convert data into clear visualisations, choose the right graph for a business question, identify basic patterns, and build a simple Power BI dashboard.

Core flow: Data → Chart → Pattern → Insight → Business Decision

1. Learning Outcomes

- Understand why data visualisation is important.
- Choose an appropriate graph for a business question.
- Create basic charts using Matplotlib.
- Create statistical charts using Seaborn.
- Interpret trends, comparisons, distributions, relationships and outliers.
- Export a CSV file from Python and import it into Power BI.
- Build a simple dashboard with a KPI card, charts and a slicer.

2. What is Data Visualisation?

Data visualisation represents data using charts and graphs so that patterns are easier to see than in a table of numbers. For example, a line chart can make a monthly sales trend immediately visible.

3. Types of Graphs – Quick Guide

Graph	Use	Question
Line chart	Trend / time	How is the value changing?
Bar chart	Category comparison	Which is bigger?
Histogram	Numerical distribution	How are values spread?
Scatter plot	Relationship	Are two numbers related?
Countplot	Category counts	How many records are in each category?
Boxplot	Spread / outliers	Are there unusual values?
Heatmap	Correlation / matrix	Which variables are more strongly related?
KPI Card	One important number	What is the total/average/value?

4. Matplotlib – Basic Visualisation

Matplotlib is a general-purpose Python plotting library. It is excellent for learning chart fundamentals and gives detailed control over a graph.

```
import matplotlib.pyplot as plt

plt.plot(x, y)      # line
plt.bar(x, y)       # bar
plt.hist(values)    # histogram
plt.scatter(x, y)   # scatter

plt.title("My Chart")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```

4.1 Line Chart

Use for trends, especially over time: monthly sales, daily temperature, yearly revenue.

```
plt.plot(df["Month"], df["Sales"], marker="o")
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

Look for: upward trend, downward trend, peaks and sudden changes.

4.2 Bar Chart

Use to compare separate categories such as sales by product or transactions by region.

```
plt.bar(df["Month"], df["Sales"])
plt.title("Sales by Month")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

4.3 Histogram

Use to understand the distribution of numerical values. Values are grouped into ranges called bins.

```
plt.hist(df["Sales"])
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
```

4.4 Scatter Plot

Use to study the relationship between two numerical variables. Each dot represents one observation.

```
plt.scatter(df["Sales"], df["Profit"])
plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()
```

5. Seaborn – What and Why?

Seaborn is a statistical visualisation library built on top of Matplotlib. It works very naturally with Pandas DataFrames and makes many statistical charts easier to create with less code.

Main advantages:

- Simple syntax for common statistical plots.
- Works directly with Pandas DataFrames.
- Useful for categorical data, distributions and relationships.
- Provides convenient charts such as countplot, boxplot and heatmap.
- Uses Matplotlib underneath, so Matplotlib commands can still be used for titles, labels and display.

```
import seaborn as sns
```

5.1 Countplot

Counts how many observations occur in each category.

```
sns.countplot(data=df, x="Region")
plt.title("Transactions by Region")
plt.show()
```

5.2 Seaborn Barplot

Compares a numerical measure across categories. With the default settings, it commonly shows an estimated central value such as the mean.

```
sns.barplot(data=df, x="Category", y="Sales")
plt.title("Average Sales by Category")
plt.show()
```

5.3 Boxplot

Summarises the spread of numerical data and can help identify possible outliers.

```
sns.boxplot(data=df, y="Sales")
plt.title("Sales Boxplot")
plt.show()
```

5.4 Heatmap

Displays a matrix visually. A common beginner use is a correlation heatmap.

```
corr = df[["Sales", "Profit"]].corr()
sns.heatmap(corr, annot=True)
plt.title("Sales and Profit Correlation")
plt.show()
```

6. Correlation – Beginner Explanation

Correlation describes the strength and direction of a linear relationship between two numerical variables. It usually ranges from -1 to +1.

Correlation	Simple meaning
Close to +1	Strong positive relationship
Around 0	Little or no linear relationship
Close to -1	Strong negative relationship

Important: Correlation does not prove causation. Two variables moving together does not automatically mean one causes the other.

7. Basic EDA + Visualisation Workflow

1. Load data
2. Inspect rows and columns
3. Check data types and missing values
4. Calculate basic statistics
5. Ask a business question
6. Choose the correct graph
7. Create the visual
8. Interpret the pattern
9. Write a business insight
10. Present it in a dashboard

8. Useful Pandas Commands

Command	Purpose
<code>df.head()</code>	First rows
<code>df.shape</code>	Rows and columns
<code>df.info()</code>	Data types / non-null information
<code>df.describe()</code>	Basic numerical statistics
<code>df.isnull().sum()</code>	Missing-value count
<code>df["Sales"].sum()</code>	Total sales
<code>df["Sales"].mean()</code>	Average sales
<code>df.to_csv("file.csv", index=False)</code>	Export to CSV

9. Power BI – Role in Day 4

Power BI is used for interactive business reporting and dashboards. In this exercise, Python prepares the data and Power BI turns the prepared data into an interactive report.

Important: Power BI Desktop is a separate application; it is not run inside Jupyter Notebook.

10. Python → CSV → Power BI

```
Jupyter Notebook
↓
Pandas DataFrame
↓
df.to_csv(...)
↓
CSV file
↓
Power BI Desktop
↓
Get Data → Text/CSV
↓
Load
↓
KPI + Charts + Slicer
↓
Interactive Dashboard
```

11. Import CSV into Power BI Desktop

1. Open Power BI Desktop.
2. Select **Home** → **Get data** → **Text/CSV**.
3. Choose the CSV generated from Jupyter, for example **day4_sales_assignment.csv**.
4. Preview the data and click **Load**.
5. The table and fields appear in the Data pane.

12. Simple Power BI Dashboard

Visual	Fields	Purpose
KPI Card	Sales	Total sales
Line chart	Month + Sales	Sales trend
Bar chart	Category + Sales	Category comparison
Bar chart	Region + Sales	Region comparison
Slicer	Region	Interactive filtering

13. KPI Card – Beginner Explanation

A KPI card displays one important business number. If you put the Sales field into a Power BI Card, Power BI can show the total sales.

```
12,000 + 15,000 + 14,000 + 18,000  
+ 21,000 + 19,000 + 23,000 + 25,000  
= 157,000
```

Depending on formatting, Power BI may display **157K** instead of 157,000.

14. Common Beginner Mistakes

- Using a line chart when there is no meaningful order or time sequence.
- Using a histogram to compare categories instead of numerical distributions.
- Forgetting chart titles and axis labels.
- Adding too many visual effects or colours.
- Showing a chart without explaining the insight.
- Treating correlation as proof of causation.
- Forgetting to sort text months chronologically in Power BI.
- Putting Sales into the wrong axis/field well.

15. Day 4 Hands-on Practice

- Create a line chart for Monthly Sales.
- Create a bar chart for Sales by Month.
- Create a histogram for Sales.
- Create a scatter plot for Sales vs Profit.
- Create a Seaborn countplot for Region.
- Create a Seaborn barplot for Category vs Sales.
- Create a boxplot for Sales.
- Create a Sales–Profit correlation heatmap.
- Create a 2 × 2 visual summary.
- Export the data to CSV and import it into Power BI.
- Build a KPI card, charts and a Region slicer.
- Write five business insights.

16. Final Revision Sheet

Pandas → Work with and prepare data.

Matplotlib → General-purpose charts and detailed control.

Seaborn → Statistical charts with concise syntax and DataFrame support.

Power BI → Interactive business reports and dashboards.

Line	→ Trend
Bar	→ Comparison
Histogram	→ Distribution
Scatter	→ Relationship
Countplot	→ Category count
Boxplot	→ Spread / outliers
Heatmap	→ Matrix / correlation
KPI Card	→ One important number

Most important rule: Start with the business question, then choose the chart. Do not choose a chart simply because it looks attractive.

17. End-of-Day Checklist

- I can explain line vs bar charts.
- I know when to use a histogram and scatter plot.
- I understand why Seaborn is useful.
- I can create basic charts in Python.
- I can export a DataFrame to CSV.
- I can import CSV into Power BI Desktop.
- I can create a KPI, chart and slicer.
- I can write a business insight from a visual.